Fine-tuning AlphaGenome on single-cell pseudobulk multiome data

1 Summary

One AlphaGenome model was fine-tuned jointly on five single-cell multiome studies (ten species-specific sources, 20 output heads). The most effective changes were to the training targets: RNA rebuilt as gene-level pseudobulk counts per million (CPM) from raw counts, ATAC rebuilt from raw fragments where released tracks were unsuitable, and removal of near-empty or unsupported cell groups. A correlation term in the loss, staged head training, and low-rank adapters gave smaller gains. The selected model reaches mean held-out chromosome correlation $R = 0.684$ for ATAC and 0.570 for RNA. On fine-mapped brain eQTLs, gene-span variant effects give AUROC 0.55 (SingleBrain) and 0.64 (XQTL), with discrimination concentrated near the TSS and only small gains from matching the eQTL cell type to its model track.

2 Data

Each study provides pseudobulk ATAC and RNA profiles for annotated cell groups (Table 1). Each source keeps its own genome assembly, and chromosomes 8 and 9 of each assembly are held out for validation and test. Macaque and marmoset sequences use AlphaGenome's human organism embedding.

Table 1: Training data. Group counts are ATAC/RNA cell groups; Liu and Johansen RNA heads have one channel per strand. ATAC source is the released track or a reconstruction from raw fragments.

Study	Tissue	Species	Groups	ATAC source
Mannens (HDA)	first-trimester brain	human	134/134	released
Liu (HDMA)	fetal, 12 organs	human	186/186	fragments
Johansen 2025	basal ganglia	human, macaque, marmoset	47/47	fragments
Zemke 2023	motor cortex	human, macaque, marmoset, mouse	20/19	released
Zemke 2024	hippocampus (aging)	human	18/18	fragments

3 Training strategies

Performance is measured by signed double-centered Pearson correlation R. For target and prediction matrices $Y, \hat{Y} \in \mathbb{R}^{N \times C}$, with N genomic elements (genes for RNA, 128 bp bins for ATAC) and C cell groups, row and column means are removed before computing the correlation, so R measures relative structure across elements and cell groups rather than assay scale.

RNA targets and loss. Released coordinate RNA tracks were replaced with gene-level targets. Raw counts are summed over the cells of each group before CPM normalization; summing per-cell normalized expression instead gives targets whose double-centered agreement with the correct ones is only 0.46–0.54 (Johansen). The model's RNA output $\hat{x} \in \mathbb{R}^{B \times C}$ over B 128 bp bins is read out per gene g as

$$\hat{y}_{g,c} = \sum_b w_{g,b}\, \hat{x}_{b,c}, \tag{1}$$

where $w_{g,b} \in [0, 1]$ is the fraction of bin b covered by exons of g, and the loss is the squared error between $\log(1 + \hat{y}_{g,c})$ and $\log(1 + y_{g,c})$. In a matched frozen-backbone comparison on Zemke 2023 that changed only this target, test RNA R rose from 0.152 to 0.421 (validation 0.250 to 0.360) with ATAC unchanged. The gene-level target also avoids a ribosomal-repeat artifact in the released marmoset tracks, where one 341 bp window carried 71% of the chromosome-9 RNA variance. Adding a per-bin coverage loss did not help (HDA, mean validation R 0.604 with versus 0.598 without), so it was dropped.

ATAC targets. Where released tracks were significance scores or poorly matched coverage, ATAC was rebuilt from raw fragments as full-depth fragment coverage per million fragments, without downsampling. For Johansen human, test R was 0.210 with released tracks and 0.458 with the rebuilt target under the same training setup.

Cell-group quality control. One Johansen group with 33,424 fragments (median 1.1×10^8) contributed 48% of all squared ATAC signal and was removed. Groups without supporting cell assignments are masked rather than given zero targets: four Zemke 2024 subtype channels and the Zemke 2023 L5 IT RNA group.

Loss. A double-centered correlation term, $1 - R$ computed within each batch, was added to the loss of every RNA head (weight 1, HDA 0.1) and of the Zemke 2023 ATAC heads (weight 3). In a matched continuation of the joint model it raised validation R from 0.568 to 0.583, mainly through RNA (0.461 to 0.491). The effect varies by study; one weighting reduced Liu RNA, which the final weights recover.

Model and optimization. The pretrained backbone is shared by all heads. Each source has its own ATAC and RNA heads, because target distributions differ between species of the same study; with equal sampling of sources this raised LoRA test R from 0.584 to 0.605 relative to one pooled head per study and modality, a comparison that changes both factors together. Heads were first trained on a frozen backbone, and the best warmup epoch (12) was used to start adapter training. Initializing heads from the pretrained neural accessibility tracks of matching cell types gave validation R 0.507 against 0.488 for random initialization at warmup epoch 4, but the selected model used random initialization. Adapters were then trained from the warmed heads: LoRA (rank 16) on attention and pair projections, plus LoCon (rank 4) on downsampling blocks 2–5, at learning rate 3×10^{-4}. Broad LoCon improved test R by 0.008 and 0.005 in two of three matched comparisons and was neutral in the third. Learning rate 10^{-3} was unstable for LoCon.

Not helpful. Higher adapter rank (LoRA 32, LoCon 8), uniform RNA loss upweighting, target value transforms, refitting heads after adapter training, and gene-balanced batch ordering did not improve held-out R.

4 Held-out chromosome results

The selected checkpoint (LoRA+LoCon, epoch 21) uses the targets above with Zemke 2024 ATAC rebuilt from fragments. Table 2 lists test R per source. Mean validation and test R are 0.585 and 0.627 (ATAC 0.665/0.684, RNA 0.504/0.570). Split-half reliability gives correlation ceilings of 0.82–0.93 for Johansen ATAC and 0.90 for Zemke 2024 ATAC, so those ATAC results are near the noise limit, whereas RNA targets have ceilings of 0.93 or higher where measured and remain well below them.

A run with released Zemke 2024 ATAC tracks and initialized heads reached test R 0.603. Most of its difference from the selected model comes from heads whose targets are identical in the two runs (for example Zemke 2023 mouse RNA, 0.311 against 0.603), so differences of a few hundredths in mean R between single runs should not be over-interpreted.

Table 2: Test-chromosome R of the selected model.

Study	Species	ATAC	RNA
HDA	human	0.817	0.660
Liu	human	0.784	0.485
Johansen 2025	human	0.644	0.584
	macaque	0.463	0.524
	marmoset	0.600	0.453
Zemke 2023	human	0.743	0.572
	macaque	0.663	0.679
	marmoset	0.615	0.649
	mouse	0.739	0.610
Zemke 2024	human	0.769	0.484
Mean		0.684	0.570

5 Fine-mapped eQTL benchmark

Positives are variants with posterior inclusion probability (PIP) above 0.75 and negatives have PIP below 0.01, within 512 kb of the gene's TSS. SingleBrain has 35 fine-mapping panels in seven brain cell classes (54,193 variants); XQTL has six cell-type panels (10,452 variants). Each variant is scored with the human Zemke 2023 RNA head on a 1 Mb window centered at the TSS. The gene-span effect is the absolute alternate-minus-reference difference in mean prediction over the gene's bins, averaged over the tracks matched to the panel's cell type. The TSS-bin summary performs worse and is not shown.

Table 3 compares the selected model with the head-only model and two adapter models that use the released Zemke 2024 ATAC tracks, and with one selected pretrained AlphaGenome track per cell type. Trained models differ by at most 0.015 in pooled AUROC, close to the variation of about 0.005 between repeated runs. The native tracks were selected on the same variants, so their values are optimistic. Discrimination is concentrated near the TSS: gene-span AUROC is 0.64–0.70 (SingleBrain) and 0.81–0.87 (XQTL) within 1 kb, and falls below 0.5 for all models beyond about 50 kb (SingleBrain) and 100 kb (XQTL) (Figure 1).

The trained tracks carry a small amount of cell-type-specific signal. Scoring each panel with every cell class's tracks, the matched class exceeds the unmatched ones by 0.024–0.032 AUROC on SingleBrain and 0.033–0.057 on XQTL, which is significant under permutation of the class assignment for every model. The effect is driven by microglia (XQTL AUROC 0.69–0.73 with

Table 3: Gene-span eQTL discrimination. AUROC and AP are pooled over panels; "within" is the mean of per-cell-class AUROC.

	SingleBrain			XQTL		
Model	AUROC	AP	Within	AUROC	AP	Within
Head-only	0.547	0.444	0.550	0.635	0.608	0.644
LoRA	0.540	0.432	0.539	0.628	0.600	0.642
LoRA+LoCon	0.540	0.434	0.542	0.637	0.615	0.643
Selected (LoRA+LoCon, raw Zemke 2024 ATAC)	0.547	0.445	0.545	0.643	0.623	0.646
Native matched tracks	0.524	0.439	0.535	0.627	0.626	0.650

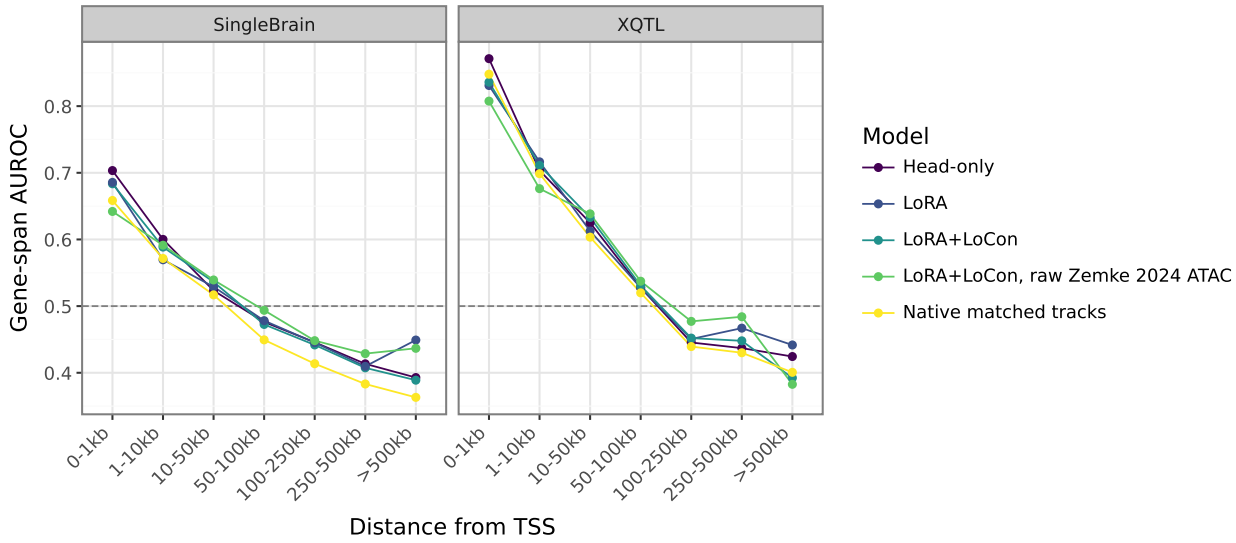


Figure 1: Gene-span AUROC by distance between variant and TSS. The dashed line is chance.

microglia tracks against at most 0.63 with other tracks), astrocytes, and oligodendrocytes; excitatory and inhibitory neuron panels show little specificity. Predicted effects are largely shared across cell types, with median Spearman correlation 0.63–0.69 between tracks.